

ENGRAM

Miner & Validator Setup Guide
theengram.space · testnet · netuid 450

Running a Miner

Step 1 — Install Prerequisites

Linux (Ubuntu 22.04+)

```
sudo apt update && sudo apt install -y \  
python3.11 python3.11-venv git docker.io  
sudo systemctl start docker
```

macOS

```
brew install python@3.11 git  
# Install Docker Desktop → docker.com/products/docker-desktop
```

Windows

```
# Open PowerShell as Administrator:  
wsl --install -d Ubuntu  
# Restart, then follow the Linux steps inside WSL
```

Step 2 — Clone & Install

```
git clone https://github.com/Dipraise1/-Engram-.git  
cd -Engram-  
python3.11 -m venv .venv  
source .venv/bin/activate # Windows: .venv\Scripts\activate  
pip install -e ".[node,qdrant]"
```

Step 3 — Start Qdrant (vector store)

```
docker run -d --name qdrant -p 6333:6333 \  
-v $(pwd)/data/qdrant:/qdrant/storage qdrant/qdrant
```

Step 4 — Configure

```
cp .env.example .env
# Then edit .env with the values below:
```

```
WALLET_NAME=default
WALLET_HOTKEY=default
SUBTENSOR_NETWORK=test
NETUID=450
MINER_PORT=8091
EXTERNAL_IP=<your-public-ip>
VECTOR_STORE_BACKEND=qdrant
USE_LOCAL_EMBEDDER=true
```

Step 5 — Create Bittensor Wallet

```
pip install bittensor

# Coldkey — save the mnemonic somewhere safe, it cannot be recovered
btcli wallet new_coldkey --wallet.name default

# Hotkey — used by the miner process
btcli wallet new_hotkey --wallet.name default --wallet.hotkey default

# Get your coldkey address
btcli wallet overview --wallet.name default
```

Step 6 — Get Testnet TAO

Go to <https://taoswap.org/testnet-faucet>, paste your coldkey address and receive test TAO (~1 min).

```
# Confirm balance arrived
btcli wallet balance --wallet.name default --subtensor.network test
```

Step 7 — Register on Subnet

```
btcli subnet register --netuid 450 \
--wallet.name default \
--wallet.hotkey default \
--subtensor.network test
```

Step 8 — Run the Miner

```
python neurons/miner.py
```

Verify it's running:

```
curl http://localhost:8091/stats
# {"status":"ok","vectors":0,"peers":5,"uid":4}
```

Step 9 — Keep Alive (auto-restart)

Linux — systemd (recommended)

```
sudo cp scripts/engram-miner.service /etc/systemd/system/  
sudo systemctl enable --now engram-miner  
journalctl -u engram-miner -f # view live logs
```

macOS / Windows (WSL)

```
pip install pm2  
pm2 start "python neurons/miner.py" --name engram-miner --cwd $(pwd)  
pm2 save && pm2 startup
```

Running a Validator

Step 1 — Clone & Install

Skip if you already ran the miner setup on this machine.

Linux (Ubuntu 22.04+)

```
sudo apt update && sudo apt install -y python3.11 python3.11-venv git
```

macOS

```
brew install python@3.11 git
```

Windows

```
wsl --install -d Ubuntu # in PowerShell as Admin, then use WSL
```

```
git clone https://github.com/Dipraise1/-Engram-.git
cd -Engram-
python3.11 -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -e ".[node]"
```

Step 2 — Generate Ground Truth Dataset

```
USE_LOCAL_EMBEDDER=true python scripts/generate_ground_truth.py --count 1000
```

*Creates **data/ground_truth.jsonl** — 1000 embedded text entries used to score miners.*

Step 3 — Configure

```
cp .env.example .env
```

```
WALLET_NAME=default
WALLET_HOTKEY=validator
SUBTENSOR_NETWORK=test
NETUID=450
GROUND_TRUTH_PATH=./data/ground_truth.jsonl
```

Step 4 — Create Bittensor Wallet

Skip if already created during miner setup.

```
pip install bittensor
btcli wallet new_coldkey --wallet.name default
btcli wallet new_hotkey --wallet.name default --wallet.hotkey validator
btcli wallet overview --wallet.name default
```

Step 5 — Get Testnet TAO

Go to <https://taoswap.org/testnet-faucet>, paste your coldkey address.

```
btcli wallet balance --wallet.name default --subtensor.network test
```

Step 6 — Register on Subnet

```
btcli subnet register --netuid 450 \  
--wallet.name default \  
--wallet.hotkey validator \  
--subtensor.network test
```

Step 7 — Run the Validator

```
python neurons/validator.py
```

Step 8 — Keep Alive (auto-restart)

Linux — systemd (recommended)

```
sudo cp scripts/engram-validator.service /etc/systemd/system/  
sudo systemctl enable --now engram-validator  
journalctl -u engram-validator -f
```

macOS / Windows (WSL)

```
pm2 start "python neurons/validator.py" --name engram-validator --cwd $(pwd)  
pm2 save
```

Scoring — validators score miners every 120s.

Component	Weight	Description
recall@10	50%	Did the miner return the expected CIDs?
Latency	30%	$100\text{ms} = 1.0 \cdot 500\text{ms} = 0.0 \cdot \text{linear}$
Proof rate	20%	HMAC challenge-response success rate

Miners below 50% proof success rate score 0 regardless of recall or latency.

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